

Qiskit v2.x Practice Quiz

Questions

1. Which of the following state vectors corresponds to the following OpenQASM 3 code?

```
OPENQASM 3;
include "stdgates.inc";

qubit[2] q;
bit[2] c;

h q[0];
cx q[1], q[0];
z q[1];
```

- A. $\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |01\rangle$
- B. $\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$
- C. $\frac{\sqrt{2}}{2} |00\rangle - \frac{\sqrt{2}}{2} |01\rangle$
- D. $\frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |11\rangle$

2. What does the following circuit prepare?

```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
```

- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

3. Given the following code fragment, what is the approximate probability that a measurement would result in a bit value of 1?

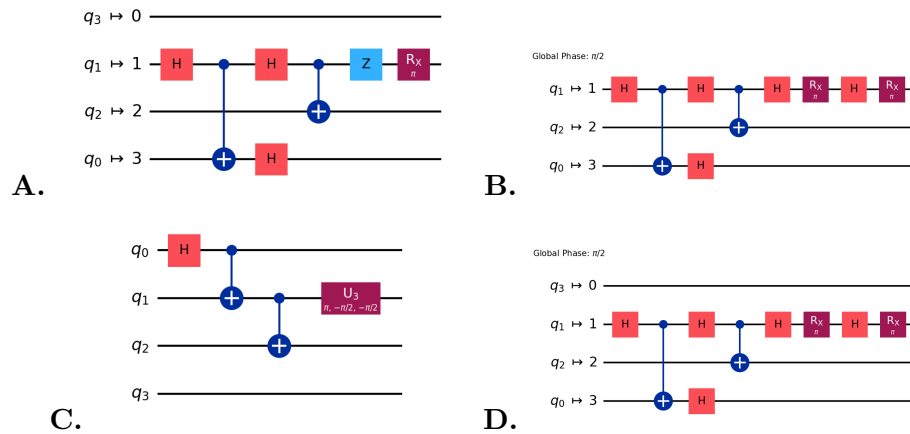
```
from qiskit import QuantumCircuit
from numpy import pi
qc = QuantumCircuit(1)
qc.reset(0)
qc.sx(0)
qc.rx(pi/2, 0)
qc.measure_all()
```

- A. 0.75
 - B. 0.0
 - C. 0.5
 - D. 1.0
4. Why may SWAP gates be inserted during transpilation?
- A. To reset qubits
 - B. To satisfy hardware connectivity constraints
 - C. To increase shot count
 - D. To remove all noise
5. What are return values of the Estimator when the observable is Pauli Z on a single qubit?
- A. 0 or 1
 - B. $|0\rangle$ or $|1\rangle$
 - C. a real number between -1 and +1
 - D. a real number between 0 and +1
6. What does layout refer to in transpilation?
- A. Mapping virtual qubits to physical qubits
 - B. Choosing the number of shots
 - C. Drawing the circuit
 - D. Choosing classical register names
7. When exporting Qiskit code to OpenQASM 3 as a file, which of these method has to be used?
- A. `export_to_qasm()`
 - B. `dumps()`
 - C. `qiskit_to_qasm()`
 - D. `dump()`
8. Which code correctly creates a two-qubit quantum circuit?
- A. `QuantumCircuit()`
 - B. `QuantumCircuit(2)`
 - C. `Circuit(2)`
 - D. `QuantumRegister.create(2)`
9. Which of the circuits below is possible output for the code below?
- ```
from qiskit import *
from qiskit_ibm_runtime import QiskitRuntimeService
from qiskit.transpiler.preset_passmanagers import (
 generate_preset_pass_manager,)
from math import pi
```

```

qc = QuantumCircuit(4)
qc.h(0)
for i in range(2):
 qc.cx(i, i+1)
qc.z(1)
qc.rx(pi, 1)
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(
 optimization_level=1,
 basis_gates=["rx", "cx", "h"],
 coupling_map=[[0, 2], [1, 3], [1, 2]]
)
qc_isa = pm.run(qc)
qc_isa.draw("mpl", idle_wires=True)

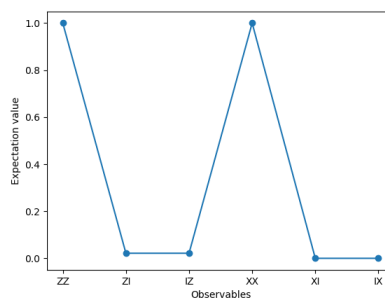
```

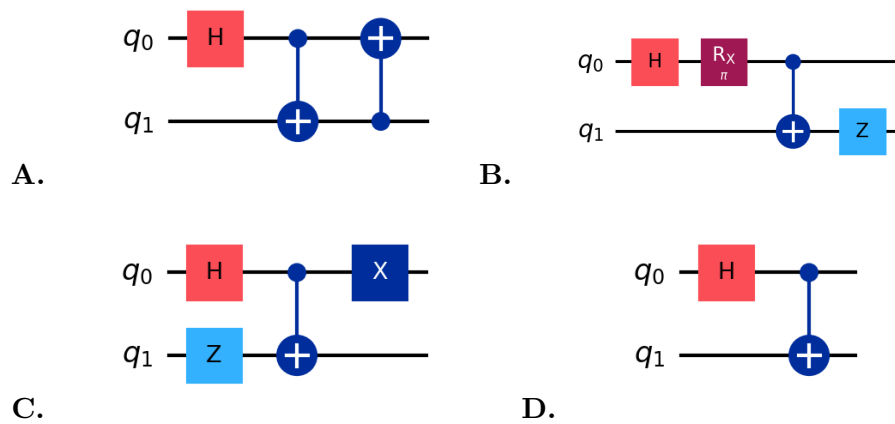


10. Which techniques are commonly configured directly through dedicated Estimator option groups rather than relying only on resilience\_level (*Select all that apply*)

- A. Dynamical decoupling
- B. Twirled readout error extinction
- C. Pauli Twirling
- D. Zero Noise Extrapolation

11. Which circuit is most consistent with the following Estimator output graph?





12. Which of the following are transpiler stages? *(Select all that apply)*

- A. Layout
- B. Routing
- C. Decomposing
- D. Scheduling

13. When is Sampler more appropriate than Estimator?

- A. When you need bitstring measurement outcomes
- B. When you need expectation values of observables
- C. When you need to draw a circuit
- D. When you need to create a backend

14. What does the Sampler primitive primarily return?

- A. Expectation values
- B. Measurement samples or bitstring results
- C. Only statevectors
- D. Only transpiled circuits

15. Which state lies on the positive X axis of the Bloch sphere?

- A.  $|0\rangle$
- B.  $|1\rangle$
- C.  $|+\rangle$
- D.  $|-\rangle$

16. What is the purpose of transpilation?

- A. To convert a circuit into backend-compatible instructions
- B. To measure all qubits automatically
- C. To create a statevector
- D. To define an observable

17. Given the code below, how would you retrieve probability distribution from sampler results? Assume all components have been imported correctly.

```
qbits = QuantumRegister(2,"qbits")
clbits = ClassicalRegister(2,"clbits")
qc = QuantumCircuit(qbits, clbits)
(q0, q1) = qbits
(c0, c1) = clbits
qc.h(0)
qc.cx(0,1)
qc.measure(qbits, clbits)
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(backend=backend, optimization_level=2)
qc_isa = pm.run(qc)
sampler = SamplerV2(mode=backend)
job = sampler.run([qc_isa])
result = job.result()
```

- A. `counts = result[0].data.meas.get_counts()`
- B. `counts = result[0].data.clbits.get_counts()`
- C. `counts = result[0].data.evs.get_counts()`
- D. `counts = result[0].data.stds.get_counts()`

18. What does increasing the number of shots usually improve?

- A. Sampling precision
- B. Circuit depth
- C. Qubit connectivity
- D. Gate fidelity automatically

19. What is the expectation value of measuring observable  $Z$  on state  $|+\rangle$

- A.  $+1$
- B.  $-1$
- C.  $+\frac{1}{2}$
- D.  $0$

20. Starting from  $|0\rangle$ , the sequence  $Rx(\frac{\pi}{2})$ ,  $Rz(\frac{\pi}{2})$ ,  $Ry(\frac{-\pi}{2})$  is applied. What single  $R(\theta, \phi)$  gate is this sequence equivalent to

- A.  $R(2\pi, 0)$
- B.  $R(\pi, \frac{\pi}{2})$
- C.  $R(\frac{3\pi}{2}, 2\pi)$
- D.  $R(\frac{\pi}{2}, 0)$

## **Answer Key**

1. S8-004: A
2. S3-002: C
3. S1-007: D
4. S4-002: B
5. S6-003: C
6. S4-003: A
7. S8-002: B
8. S3-001: B
9. S3-006: D
10. S6-006: A, C
11. S2-004: D
12. S3-007: A, B, D
13. S5-002: A
14. S5-001: B
15. S2-002: C
16. S4-001: A
17. S7-004: B
18. S7-001: A
19. S1-005: D
20. S1-006: A